

Lekkala Ganesh

AI/ML Engineer | LLM Applications, RAG & Data Science

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SUMMARY

AI/ML engineer building LLM applications and data-science models in Python: RAG pipelines (chunking, embeddings, vector search), conversational AI — chatbots and voicebots with speech-to-text and text-to-speech — LangGraph agents with guardrails, and the REST APIs behind them, shipped with statistical evaluation and structured logging. Automation Testing (500+ automated tests) and CI across public repos. B.Tech CSE.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, HTML5, CSS3

Machine Learning & Statistics: scikit-learn, Pandas, NumPy, SciPy, Regression, Classification, Feature Engineering, Cross-Validation, Hyperparameter Tuning, Model Evaluation, EDA, TF-IDF, Cohen's Kappa, Wilcoxon Signed-Rank, Class Imbalance, Precision/Recall

Generative AI & Agents: GPT-4o, Gemini, Ollama (local inference), Hugging Face Transformers, PyTorch, LangChain, LangGraph, MCP, Tool Calling, Guardrails, Prompt Engineering, Multi-Provider Fallback, SSE Streaming

RAG & Conversational AI: Embeddings (sentence-transformers), Chunking, Vector Search (LanceDB, pgvector, ChromaDB), Semantic Search, Chatbots, Voicebots, Speech-to-Text (Whisper, Web Speech API), Text-to-Speech, Intent Classification

Backend & Data: FastAPI, Flask, REST APIs, OpenAPI/Swagger, JSON Schema Validation (Pydantic, Zod), WebSockets, MERN (MongoDB, Express.js, React, Node.js), PostgreSQL (Row-Level Security, Triggers, Indexes), SQLite, Supabase, Schema Design, Joins & Aggregations

Engineering & Ops: Automation Testing, pytest, Vitest, Playwright, GitHub Actions (CI), Docker, Git, Linux, Structured Logging, Correlation IDs, Token-Cost Tracking, ONNX Runtime, Vercel

EXPERIENCE

Freelance AI/ML & Full-Stack Engineer

Remote

Self-Employed

Jan 2025 – Present

- **Client Websites:** Designed and developed **2 responsive, accessible** business websites end-to-end (**UI/UX**, semantic **HTML5/CSS3**, modern **CSS Grid/Flexbox**, cross-browser and mobile-first layout), optimized for **Core Web Vitals** (LCP, CLS) and **on-page SEO** with conversion-focused, lead-generation-ready layouts, for **U.S.-based** construction-industry clients — [Ultracore Construction](#) and [McAllister Builders](#).
- **Client Communication:** Ran requirement-gathering, revision, and sign-off calls directly with **U.S.-based** clients across time zones, translating verbal feedback into iterative design/copy changes with same-day turnaround.
- **Marketplace Platform (Ongoing):** Building Aglaowner, a **Next.js** self-serve small-business marketplace — mobile **OTP** and **DigiLocker** identity verification, **WhatsApp**-based contact unlock, listing search/filter across 800+ live listings.
- **Fintech Education Platform (Completed):** Built [IQCommune](#), a **Next.js** site for a financial-literacy training platform with interactive **budgeting**, **retirement**, and **portfolio-valuation** calculators and a waitlist signup flow.

Junior Software Developer

Mumbai, India

AlterSquare (OrderStack Partnership)

Dec 2025 – Mar 2026

- Designed and developed the **responsive, accessible** client website [2A Construction](#) end-to-end (**UI/UX**, semantic **HTML5/CSS3**, **CSS Grid/Flexbox**, mobile-first, cross-browser layout) for a **U.S.-based** client, optimized for **Core Web Vitals** and **on-page SEO**.

Web Developer

Ahmedabad, Gujarat (On-site)

Dash Technologies Inc.

Jan 2025 – Jul 2025

- Built **12+ modular React** components for a patient-facing **healthcare portal**, cutting **accessibility audit findings** by **40%** via **semantic HTML** and **HIPAA-aligned** data handling.
- Achieved **100% cross-browser visual parity** (Chrome, Firefox, Safari, Edge) for the **appointment scheduling** and **patient history** modules.

Full Stack Web Development Intern

Remote, Part-time

Slash Mark IT Solutions

Dec 2024 – Apr 2025

- Delivered **6 full-stack projects** on the **MERN stack** (MongoDB, Express.js, React, Node) — including a URL Shortener **REST API** and a Food Delivery app with **authentication**, geolocation and payment flow.

Machine Learning Intern

Virtual

Bharat Intern

Oct 2023 – Nov 2023

- Built **regression** and **classification** models in **Python** with **scikit-learn** and **Pandas** — **data cleaning**, **EDA**, **feature engineering**; compared estimators on held-out metrics via **cross-validation** and **hyperparameter tuning**.

PROJECTS

- Lenny Growth Assistant** | *Python, FastAPI, pgvector, BM25, Next.js, Docker* [GitHub](#)
- Grounded **RAG** over podcast transcripts: **hybrid retrieval** (pgvector **HNSW** + **BM25**, **Reciprocal Rank Fusion**) with episode/timestamp **citations** and out-of-scope refusal; **multi-LLM fallback** (OpenAI → Groq → Gemini, Ollama, Claude) and sandboxed artifacts.
- AI Interview Prep Kit** | *TypeScript, Node.js, LLM APIs, CI* [GitHub](#)
- Turns a job description and company site into a **requirement-traced** interview kit; **structured JD extraction**, a **deterministic coverage engine** with bounded gap-filling pass, N-day study plan, flashcards, and 5 **LLM providers**.
- Agentic AI RAG Chatbot** | *Python, LangGraph, Pinecone, OpenAI, FastAPI* [GitHub](#)
- Document-grounded chatbot: PDF chunking → **OpenAI embeddings** → **Pinecone** → **LangGraph** relevance gate that **refuses below a retrieval-confidence threshold**, served via **FastAPI**.
- Zoho One Employee Portal** | *Next.js 16, Express, TypeScript, Supabase Postgres* [GitHub](#)
- Portal with its own **JWT** (httpOnly, revocable sessions) and **RBAC**; employees reach **Zoho CRM/Desk/Books/People** through one server-held service account, with **audit logs** and **RLS**-locked tables.
- Hotel Bar Inventory Intelligence** | *Python, pandas, statsmodels, scikit-learn* [GitHub](#)
- **Demand forecasting** (Holt-Winters, RandomForest) and dynamic **par levels** with safety stock across 96 bar-brand series; backtest cut **stockout days 73–87%** and lifted **fill rate 28% → 94%**.
- AI Study Assistant** | *React, TypeScript, Gemini API, Structured Output* [GitHub](#)
- Turns notes into **flashcards** and a **quiz** via a **server-side Gemini** call with **structured JSON output**; **runtime validation** before render, **request cancellation** and stale-response guards, full loading/error/retry states.
- Cost-Efficient RAG Service** | *Python, FastAPI, LanceDB, sentence-transformers, litellm* [GitHub](#)
- **Chunking** → **embeddings** (all-MiniLM-L6-v2) → **LanceDB vector search** → generation, with a second **pgvector** backend; **0.71 Recall@5**, **34.6 ms p95**, documented **token-cost/TCO** comparison.
- LLM-as-Judge Evaluation Pipeline** | *Python, scikit-learn, SciPy, Rubric Design* [GitHub](#)
- Measures 5 judge biases (position, verbosity, self-enhancement, sycophancy, score clustering) using **Cohen’s kappa** and **Wilcoxon signed-rank** tests against **105 hand-labeled** human judgments; **200 tests**.
- ATS Resume Optimizer** | *Python, scikit-learn, Streamlit, GitHub Actions* [GitHub](#)
- Resume-to-job **semantic matching** via **scikit-learn TF-IDF** and **cosine similarity** with keyword-gap extraction; **276 pytest tests** and **GitHub Actions CI** (3.11/3.12, ruff) over **~20K lines of Python**.

EDUCATION

Parul University	Vadodara, Gujarat
<i>Bachelor of Technology in Computer Science & Engineering; GPA: 7.53/10</i>	<i>Jul 2022 – Jun 2025</i>